



Strategy and Innovation stream

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Mansi

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FX risk management policy for a leading technology firm headquartered in India

OBJECTIVES	DESCRIPTION
RISK ASSESSMENT	Identify potential foreign exchange risks, including currency fluctuations.
REGULATORY COMPLIANCE	Ensure adherence to guidelines set by RBI and FEMA.
RISK MITIGATION STRATEGIES	Develop strategies to minimize the impact of currency volatility.
HEDGING INSTRUMENTS	Explore and implement diverse hedging options, such as forwards and options.
MONITORING AND REPORTING	Establish a continuous review process for policy effectiveness.

Key Questions

What are the primary foreign exchange risks faced by the technology firm?

FX Risk Assessment Framework

- Identify transactional, translational, and economic risks.
- Align with international standards.

1

What regulatory frameworks and compliance requirements must be adhered to in managing FX exposure?

Continuous Monitoring Systems

- Implement real-time FX data tracking.
- Provide regular reports to senior management for informed decision-making

3

How can the firm effectively assess and mitigate these risks?

Risk Mitigation Strategy

- Utilize hedging instruments:
 - Forward contracts
 - Options
 - Future contracts
- Tailor strategies to firm's exposure profile.

2

What monitoring and reporting processes will ensure ongoing risk management effectiveness?

Regulatory Compliance

- Ensure adherence to RBI and FEMA guidelines.
- Conduct regular audits and update policies per regulatory changes.

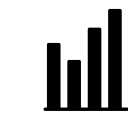
4

Key Performance Indicators (KPIs):



FX EXPOSURE MEASUREMENT

Net FX Position: Total net position across currencies to gauge exposure.



HEDGING EFFECTIVENESS

Hedging Ratio: Portion of exposure hedged vs. total exposure.



COST OF HEDGING

Hedging Cost as % of Revenue: Evaluates cost-efficiency relative to revenue.



SCENARIO ANALYSIS

Impact Forecasting: Projects effects of extreme FX movements on cash flows and profitability.



VALUE AT RISK (VaR)

Daily VaR: Estimates potential losses in normal market conditions over time.

Understanding Core Operations



Software Development & Delivery

The company provides software development services to clients worldwide, often billing in different currencies. This creates exposure to currency fluctuations, which can impact the revenue when converted back to the company's reporting currency.



Cloud Services Revenue

Offers cloud services internationally, where revenues are collected and costs are incurred in various currencies. Currency changes can affect operating margins if revenues or costs shift unfavorably.



Global IT Consulting

Provides consulting services globally, earning fees in multiple local currencies. Currency fluctuations can impact the financial statements when these earnings are consolidated into the firm's reporting currency.



Digital Transformation Projects

Manages large digital transformation projects that involve payments in various currencies. These multi-currency payments can lead to risks impacting the profitability of the projects.



Offshore Development Centers

Operates development centers in different countries, with costs incurred in local currencies while revenues may be earned in other currencies. Exchange rate changes can affect the overall cost structure.



SaaS Subscriptions

Earns revenue from SaaS subscriptions worldwide, with foreign subsidiaries generating income in different currencies. This exposes the company to translation risks during financial consolidation.



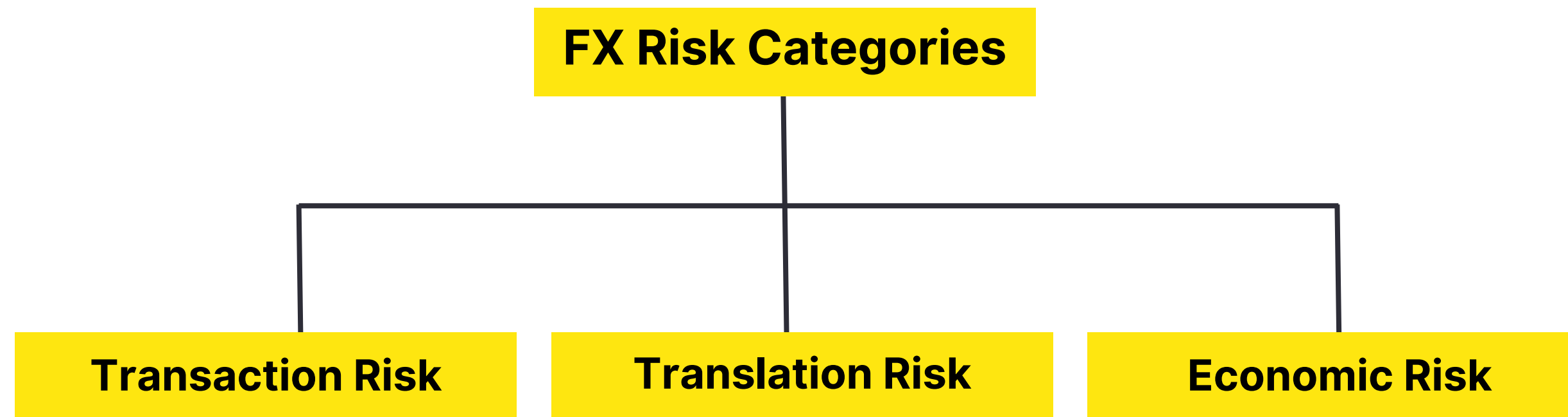
M&A Activities

Involved in cross-border mergers and acquisitions, where valuations of assets and liabilities are done in foreign currencies. Currency fluctuations can affect the value of investments and strategic growth decisions.

Risk Assessment



The purpose of the risk assessment in FX risk management is to systematically identify, measure, and analyze the potential impacts of currency fluctuations on various business operations. This helps the company understand its exposure to foreign exchange risks and develop strategies to minimize financial losses, protect profit margins, and ensure overall financial stability in a global market.



Transaction Risk

- Definition: Arises from exchange rate fluctuations affecting the value of receivables or payables in foreign currencies. It occurs between the time a transaction is initiated and settled.
- Example: A company exports goods to a foreign market and invoices in the customer's currency. If the customer's currency depreciates before payment, the company's revenue in the home currency decreases.
- Impact Areas: Affects cash flows, profit margins, and working capital.
- Mitigation Strategies: Forward contracts, options, or invoice in the home currency to lock in exchange rates and reduce uncertainty.
- Assessment Techniques: Analyze upcoming foreign currency payments and receipts to quantify exposure and identify potential financial impact.

Translation Risk

- Definition: Occurs when consolidating financial statements of foreign operations into the reporting currency. Exchange rate changes can impact the value of foreign assets, liabilities, and earnings.
- Example: A company with a subsidiary in another country sees the subsidiary's earnings reported in a weaker currency, resulting in lower consolidated profit in the home currency.
- Impact Areas: Affects reported net income, balance sheet values, and key financial ratios.
- Mitigation Strategies: Balance sheet hedging or matching assets and liabilities in the same currency to minimize the impact.
- Assessment Techniques: Evaluate the currency composition of foreign subsidiaries' assets and liabilities and simulate the impact of currency changes on financial statements.

Economic Risk

- Definition: Refers to the long-term impact of currency changes on a company's market value and competitiveness, affecting future cash flows and profitability.
- Example: A strengthening home currency makes the company's products more expensive in international markets, reducing foreign sales and market share.
- Impact Areas: Influences the company's competitive position, future revenues, cost structure, and long-term growth.
- Mitigation Strategies: Diversify operations geographically, natural hedging by matching revenues and expenses in the same currency, or relocating production to other regions.
- Assessment Techniques: Analyze long-term market trends, perform sensitivity analysis on currency impacts, and forecast future cash flows considering potential currency fluctuations.

Risk Assessment by Operation of Technological Firms

Evaluating the foreign exchange risks associated with different business activities. It identifies how specific operational processes are exposed to currency fluctuations. This approach allows for a targeted analysis of potential impacts on financial performance, enabling the company to implement appropriate risk mitigation strategies for each operational area.

Business Process	Risk Type	Exposure	Impact Area
Software Development & Delivery	Transaction Risk	Billing clients in different currencies	Revenue
Cloud Services Revenue	Economic Risk	Operating margin variability from currency fluctuations.	Operating Margins
Global IT Consulting	Translation Risk	Consulting Fees	Financial Statements
Digital Transformation Projects	Transaction Risk	Multi-currency project payments	Project Profitability
Offshore Development Centers	Economic Risk	Operating cost variations	Cost Structure
SaaS Subscriptions	Translation Risk	Foreign subsidiary consolidation	Balance Sheet
M&A Activities	Economic Risk	Cross-border valuations	Strategic Growth

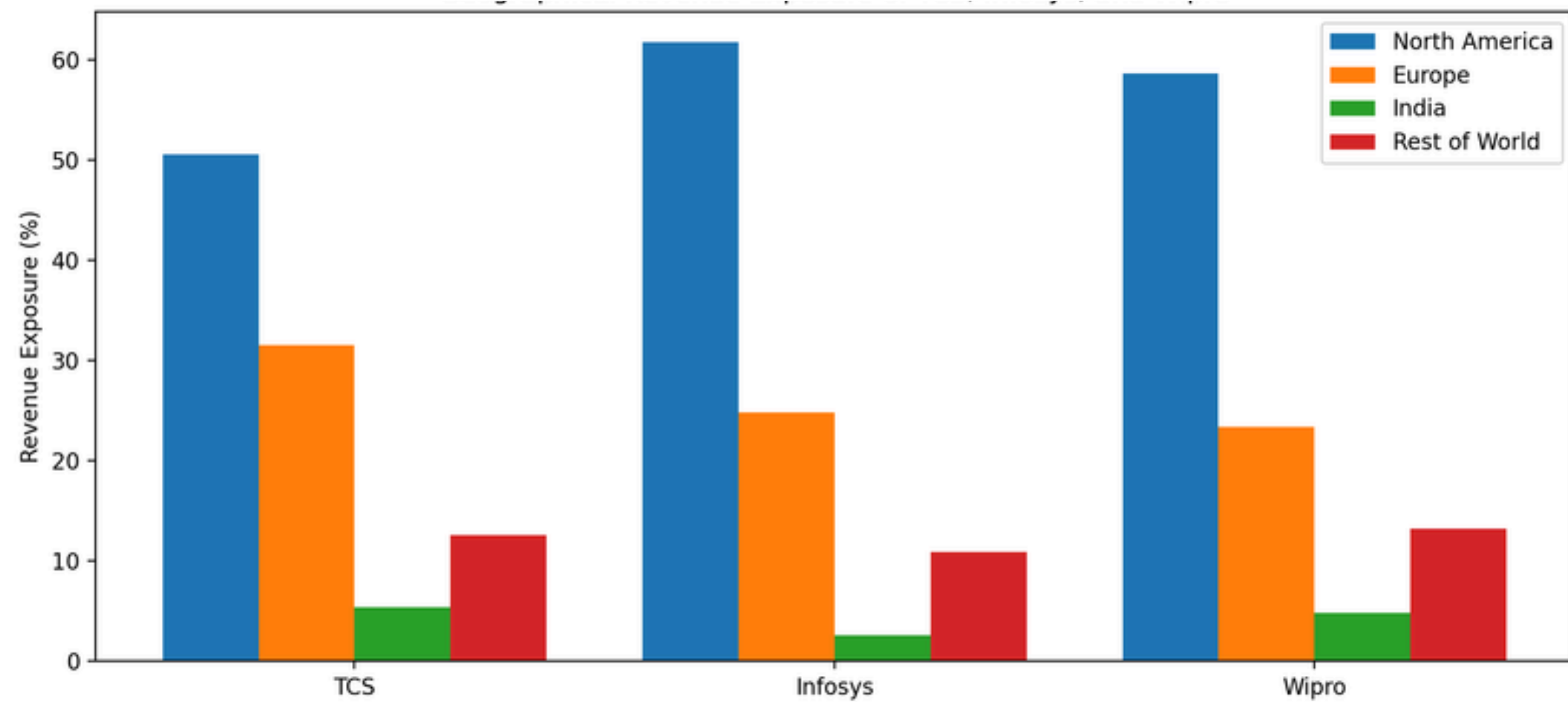
Risk Assessment



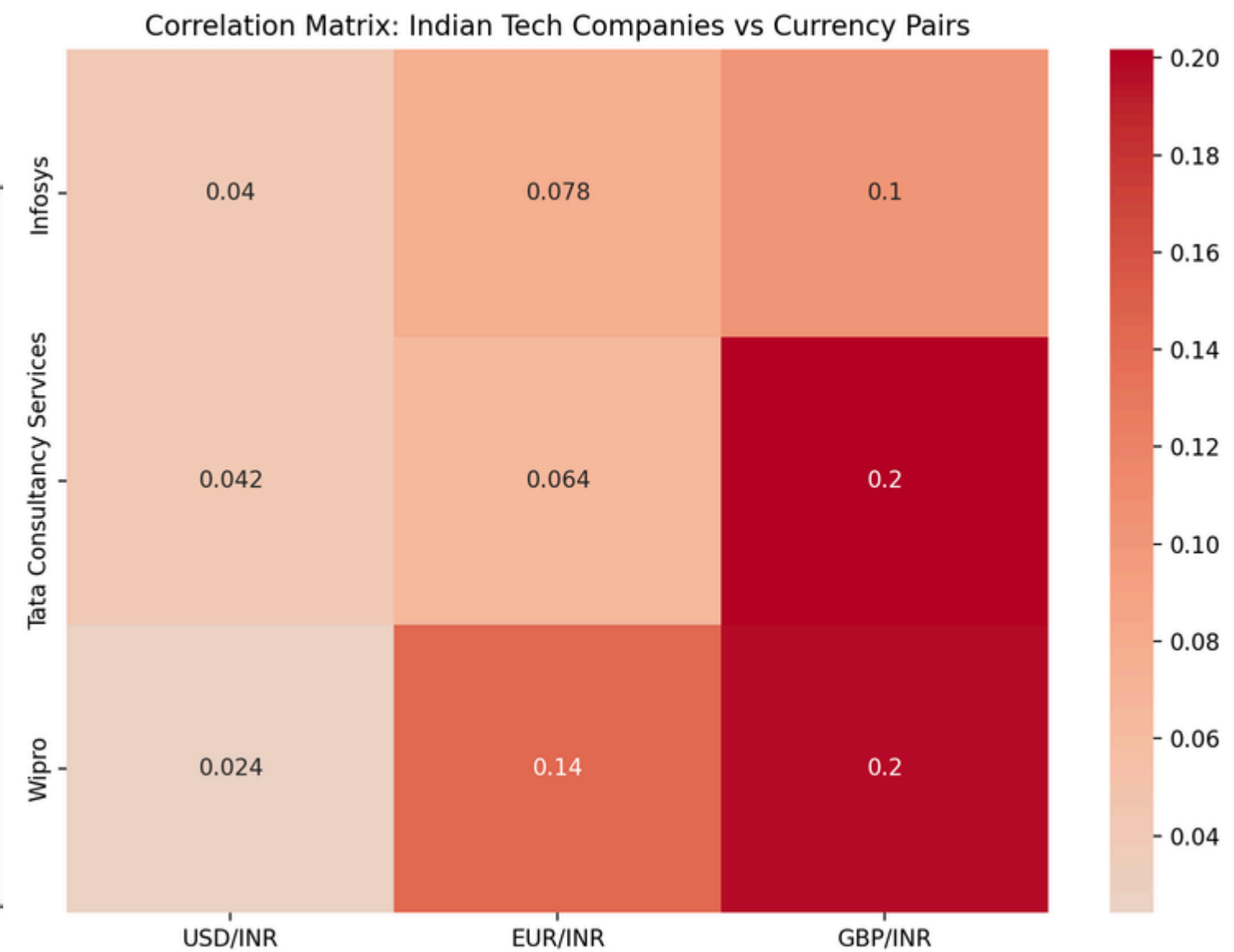
Quantitative Analysis

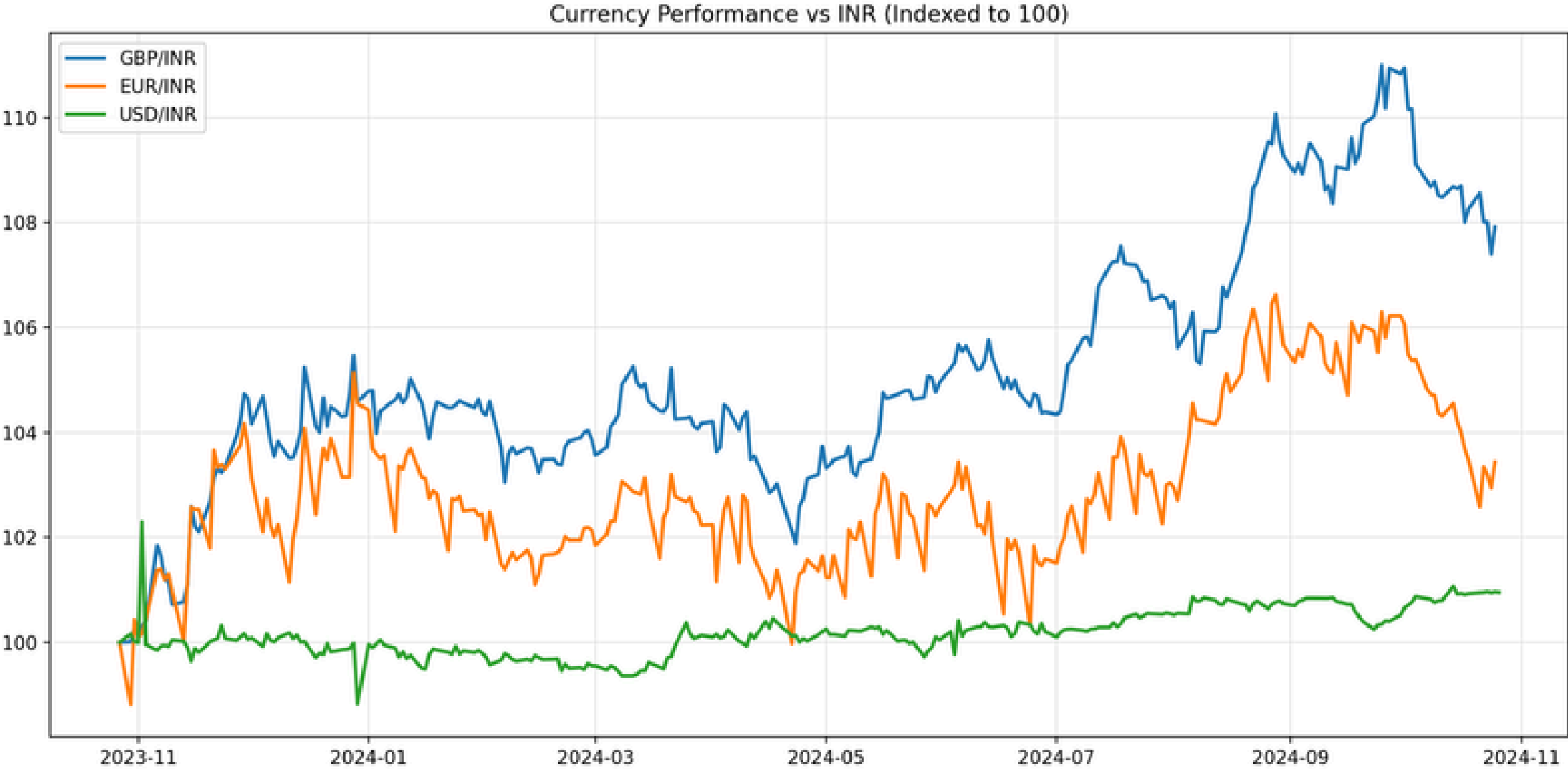
Since the technology firm is headquartered in India, the analysis focuses on currency pairs relevant to the primary markets where Indian tech companies frequently conduct business. These major outsourcing zones include regions such as the United States, Europe, and the United Kingdom, where the firm generates substantial revenue and incurs costs. Therefore, currency pairs like USD/INR, EUR/INR, and GBP/INR are considered crucial for assessing foreign exchange risk exposure, reflecting the firm's significant financial interactions in these key markets.

Major Tech company specific currency exposure for FY2023



(Source - Annual Report FX Management TCS, Infosys, Wipro)





(Source - Bloomberg)

Volatility(Daily Returns):	
GBP/INR:	0.38%
EUR/INR:	0.6%
USD/INR:	0.25%

Risk Assessment



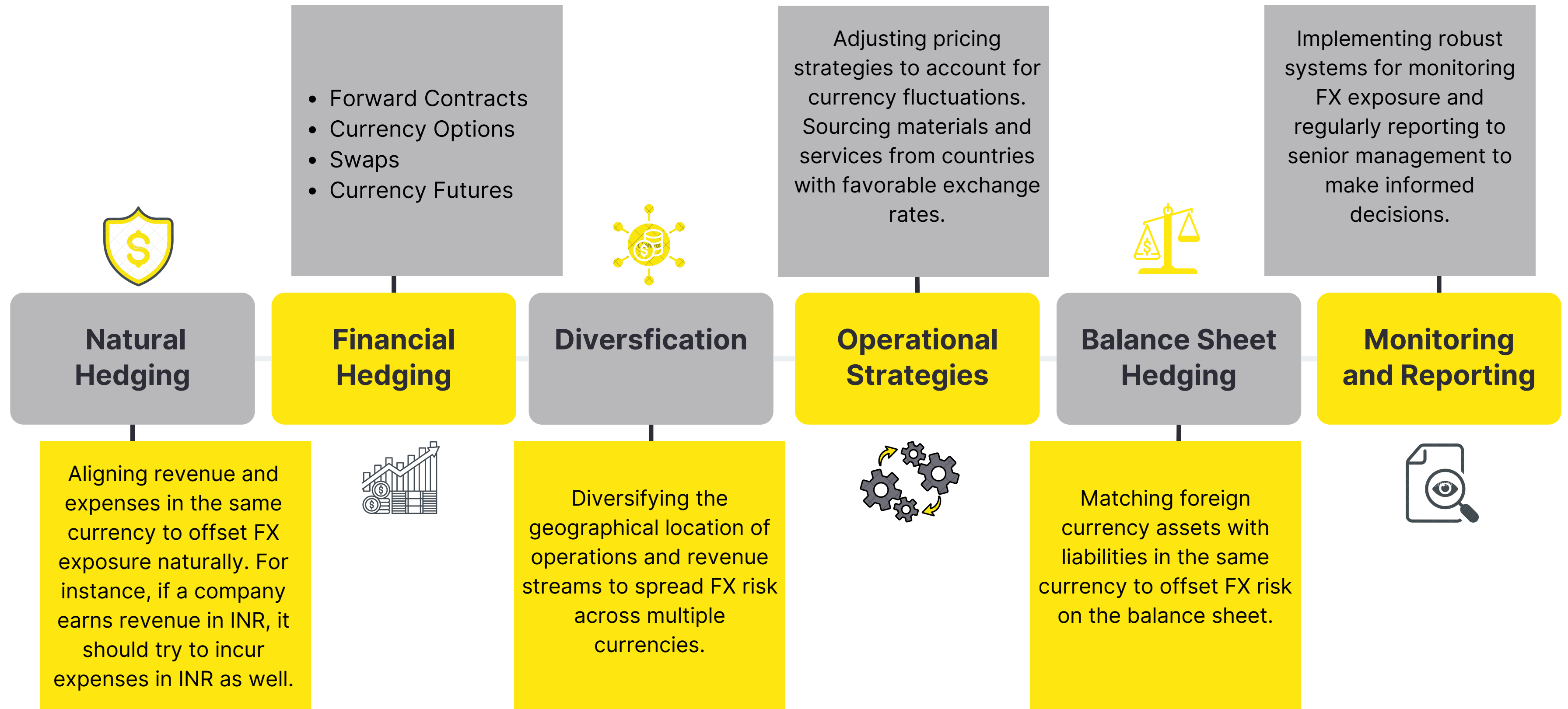
Value at Risk(95% Confidence interval)

Currency	Average Rate	Annual Volatility(%)	VaR 95% Daily	Monthly Revenue	Monthly INR Exposure	Net Monthly Exposure
USD	82.57708667	4.692440901	-0.003060126	500	50000	-8711.456666
EUR	89.1475093	9.638127807	-0.010049516	300	50000	-23255.74721
GBP	102.6971345	7.701435225	-0.00740274	200	50000	-29460.5731

Exposure Analysis

Month	Currency	Exchange Rate	Monthly Volatility(%)	Net Exposure Million (INR)
31-08-2024	USD	83.91047478	0.348186531	41955.23739
30-09-2024	USD	83.80943371	0.348186531	41904.71686
31-08-2024	EUR	93.36800003	1.972335966	28010.40001
30-09-2024	EUR	92.91704777	1.972335966	27875.11433
31-08-2024	GBP	110.7697411	1.895276296	22153.94821
30-09-2024	GBP	110.68714	1.895276296	22137.42799

Risk Mitigation Strategy



Hedging Instruments



Hedging instruments are financial tools used to mitigate the risks associated with foreign exchange fluctuations. These instruments help stabilize cash flows, protect profit margins, and ensure predictable financial performance amid currency volatility. By employing hedging instruments such as forward contracts, currency options, swaps, and futures, the company can effectively manage its exposure to foreign currencies resulting from international operations, transactions, and investments. This approach supports strategic planning and enhances the company's competitiveness in the global market.

Instrument	Description	Benefits	Drawbacks	Formula
Forward Contracts	Agreement to buy/sell currency at a fixed rate on a future date	Provides certainty on future cash flows	No flexibility if exchange rates improve	$\text{Forward Rate} = \text{Spot Rate} \times (1 + \text{rate domestic}) / (1 + \text{rate foreign})$
Options	Gives the right, but not the obligation, to buy/sell currency at a specified rate	Offers flexibility; can benefit from favorable movements	Premium costs may be high	$\text{Option Premium} = \text{Spot Rate} \times e^{(-r \times t)} \times N(d1) - \text{Strike Price} \times e^{(-r \times t)} \times N(d2)$
Currency Swaps	Exchange of principal and interest payments in different currencies	Helps manage long-term exposure and financing costs	Complexity in structuring agreements	$\text{Swap Value} = (\text{PV of Foreign Currency Payments}) - (\text{PV of Domestic Currency Payments})$
Futures Contracts	Standardized contracts traded on exchanges to buy/sell currency at a set price	Transparency and liquidity	Limited to certain currencies and contract sizes	$\text{Futures Price} = \text{Spot Rate} \times (1 + \text{rate domestic} - \text{rate foreign}) \times t$
Natural Hedging	Matching revenues and expenses in the same currency	Low cost; reduces need for financial instruments	May not fully cover all exposures	N/A – relies on balancing currency inflows and outflows

Hedging Instruments



Different hedging tactics

Hedging Tactic	Description	When to Use	Practical Use Case
Hedge Ratio	Proportion of the exposure that is hedged using financial instruments.	Use for partial hedging, dynamic adjustments, or optimizing portfolio hedging.	A technology firm hedging 70% of its USD exposure to reduce risk while allowing some upside.
Vanilla Put Options	Gives the right to sell a currency at a set price before a specified date.	Use for uncertain cash flows, downside protection, or flexible hedging strategies.	A company concerned about potential EUR depreciation buys a put on EUR/INR to cap losses.
Forward Contracts	Agreement to exchange currencies at a fixed rate on a future date.	Use for fixed, known future cash flows where certainty is desired.	A firm with an upcoming USD payment locks in the exchange rate to avoid future volatility.
Natural Hedging	Aligning currency inflows with outflows to reduce exposure.	Use when matching revenues and expenses in the same currency is feasible.	An Indian company with US-based revenues pays its US suppliers in USD, naturally offsetting FX risk.
Dynamic Hedging	Adjusting hedge positions based on market conditions or exposure changes.	Use when market volatility is high or exposure levels fluctuate frequently.	A company regularly reassesses its hedge ratio based on the volatility of the USD/INR exchange rate.
Cross-Currency Swaps	Exchange of principal and interest payments in different currencies over time.	Use for managing long-term currency exposures, especially in debt financing.	A firm borrowing in USD swaps its interest payments to INR to avoid FX risk on its debt.

Options Collar	Combination of a call and a put to limit potential gains and losses.	Use when a company wants to hedge risk while still benefiting from limited favorable currency movements.	A company hedges EUR/INR by buying a put and selling a call at different strike prices.
Currency Diversification	Spreading currency exposure across multiple currencies to reduce overall risk.	Use when revenues are generated from multiple countries with different currencies.	A firm with clients in Europe, the US, and the UK diversifies its cash holdings across EUR, USD, and GBP.
Operational Hedging	Shifting business operations or sourcing to regions with favorable currency trends.	Use when strategic changes in supply chain or location are possible.	A technology firm relocates some production from Europe to Asia to take advantage of favorable currency rates.
Stop-Loss Orders	Setting a predetermined exchange rate at which to buy or sell a currency automatically.	Use to limit potential losses if a currency moves beyond a specified threshold.	A company sets a stop-loss on USD/INR to protect itself if the rupee depreciates rapidly.

Hedging Instruments

Strategies

- **Forward Contracts**

Most General Hedge

- TCS and Infosys typically hedge 60-80% of their expected revenue using forward contracts.

Description	TCS Values
Expected Revenue (USD)	100000000
Hedged Amount (USD)	80000000
Forward Rate (INR/USD)	85.69
Guaranteed Revenue (INR)	6855200000
Revenue at Spot Rate (INR)	6725600000
Net Benefit from Hedging (INR)	129600000
Hedging Cost (INR)	250000000
Effective Rate (INR)	85.69
Cost-Benefit Ratio	1.93
Savings Percentage	0.52

Guaranteed Revenue=80,000,000×85.69=₹6,855,200,000

Calculation for Spot Rate:

Spot Rate: ₹84.07

Revenue at Spot=80,000,000×84.07=₹6,725,600,000

- **Hedging Cost Analysis**

Assumption: Assume average hedging costs of 0.8-1% of revenue.

For TCS (Annual Revenue approx \$25 billion):

Hedging Cost:

Hedging Cost=0.01×25,000,000,000=₹250,000,000

Effective Rate Calculation:

Effective Rate:

Effective Rate=Guaranteed Revenue/Hedged

Amount=6,855,200,000/80,000,000=₹85.69

TCS has an effectiveness of 85%, the hedging would protect 85% of the currency fluctuation risks

- **Savings from Hedging as a Percentage of Revenue**

For TCS:

Savings Percentage=

(129,600,000/25,000,000,000)×100≈0.52%

- **Cost-Benefit Ratio=Hedging Cost/Net Benefit**

Cost-Benefit Ratio=250,000,000/129,600,000≈1.93:1

Hedging Instruments

- **Options Strategy**

Usage:

Companies like Wipro use a combination of options for flexibility.

Typical Structure:

Buy Put Options: For downside protection

Sell Call Options: To reduce premium costs

Description	Values
Put Strike (INR/USD)	83
Call Strike (INR/USD)	85
Net Premium (INR/USD)	0.45
Protection Cost (INR)	2,25,00,000

short-term time period, such as a 3-month options strategy.

Net Premium: ₹0.45 per USD

Exposure Amount (Notional): \$50 million

The protection cost would be:

Protection Cost=Net Premium×Notional Amount

Cost=0.45×50,000,000=₹22.5million

- **Optimal Hedge Ratio**

Current Market:

70-75% of exposure based on volatility and cost metrics

Higher ratio for near-term exposure

- **Natural Hedge**

Strategy:

Companies maintain USD expenses that account for 20-30% of revenue to offset currency exposure.

Description	Values (USD)
Revenue	1,00,00,00,000
USD Expenses	25,00,00,000
Net Exposure	75,00,00,000
Actual Hedging Needed	60,00,00,000

- **Rolling Hedge Program Structure**

Layering hedges over time:

Quarter 1: 70-80% hedged

Quarter 2: 40-50% hedged

Quarter 3: 20-30% hedged

Quarter 4: 0-10% hedged

- **Cost Structure Average Hedging Costs**

Forward Points: 1.5-2% annualized

Option Premium: 1-1.5% of notional amount

Total Impact: 0.8-1.2% of revenue

- **Financial Impact**

Average Savings: 1.5-2% of revenue

Cost of Hedging: 0.8-1% of revenue

Net Benefit: 0.7-1% of revenue

Hedging Instruments



Hypothetical Case Study

The hypothetical case study of XYZ Technologies to illustrate the application of hedging instruments to effectively manage foreign exchange risk in a technology firm with headquarters in India.

XYZ Technologies is a leading Indian multinational corporation headquartered in Bangalore, specializing in software development, cloud computing, and IT consulting services. With a diverse portfolio serving clients across the USA, Europe, and the UK, the company generates significant revenue from international operations, predominantly in USD and EUR. To mitigate the impact of foreign exchange (FX) fluctuations on its profitability, XYZ Technologies employs a comprehensive FX risk management strategy.

FX Exposure and Hedging Analysis				
Category	India (INR Crore)	USA (USD Million)	Europe (EUR Million)	UK (GBP Million)
Net Exports/Imports	10,680 - 3,580	13,400 - 1,800	9,800 - 1,260	4,500 - 610
Net Exposure	7,100	11,600	8,540	3,890

Financial Overview for FY 2023

Category	India (INR Crore)	USA (USD Million)	Europe (EUR Million)	UK (GBP Million)
Revenue from Operations	17,800	13,400	9,800	4,500
- Domestic Sales	7,120			
- International Sales	10,680	13,400	9,800	4,500
Costs				
- Cost of Material Consumed (Domestic)	4,900			
- Cost of Material Consumed (Imports)	3,580	1,800	1,260	610
- Employee Cost	2,140	1,000	1,850	480
- Other Expenses	5,200	2,900	2,470	900
Total Cost	15,820	5,700	5,580	1,990
Profit Before Tax	1,980	7,700	4,220	2,510
Estimated Tax Rate	30%	21%	25%	19%
Tax	594	1,617	1,055	477
Profit After Tax	1,386	6,083	3,165	2,033

Hedging Instruments

Hedging Strategies Against USD/INR Currency Pair

Hedging Strategy	Description	Assumed Exposure (USD Million)	Hedge Ratio (%)	Hedged Amount (USD Million)	Protection Cost (INR Crore)	Guaranteed Revenue (INR Crore)	Net Benefit (INR Crore)
Forward Contracts	Locks in exchange rate for future revenue from USD sales.	11,600	70%	8,120	22.86	6,810.80	6,787.94
Options Strategy	Buy puts for downside protection; sell calls to reduce premium costs.	11,600	20%	2,320	10.44	1,929.60	1,919.16
Natural Hedge	Maintain USD expenses to offset revenue exposure.	11,600	30% of revenue				

Forward Contracts:

Hedged Amount: $11,600 \times 0.70 = 8,120$ million USD
 Protection Cost: $8,120 \times 0.45 = 22.86$ crore INR
 Guaranteed Revenue: $8,120 \times 84 = 6,810.80$ crore INR
 Net Benefit: $6,810.80 - 22.86 = 6,787.94$ crore INR

Options Strategy:

Hedged Amount: $11,600 \times 0.20 = 2,320$ million USD
 Protection Cost: $2,320 \times 0.45 = 10.44$ crore INR
 Guaranteed Revenue:
 Assuming the protection of USD at Put Strike:
 $2,320 \times 83 = 1,929.60$ crore INR
 Net Benefit: $1,929.60 - 10.44 = 1,919.16$ crore INR

Total Outcomes Overview:

Total Protection Cost (Forward + Options): $22.86 + 10.44 = 33.30$ crore INR
 Total Guaranteed Revenue (Forward + Options): $6,810.80 + 1,929.60 = 8,740.40$ crore INR
 Total Net Benefit (Forward + Options): $6,787.94 + 1,919.16 = 8,707.10$ crore INR

In conclusion, the hypothetical case study of XYZ Technologies underscores the critical importance of implementing effective foreign exchange risk management strategies. By utilizing various hedging instruments, including forward contracts, options, and natural hedges, the company can mitigate the impacts of currency fluctuations on its revenues and costs. The analysis highlights the financial outcomes of these strategies, demonstrating potential benefits such as enhanced profitability and reduced volatility in financial performance

Hedging Instruments

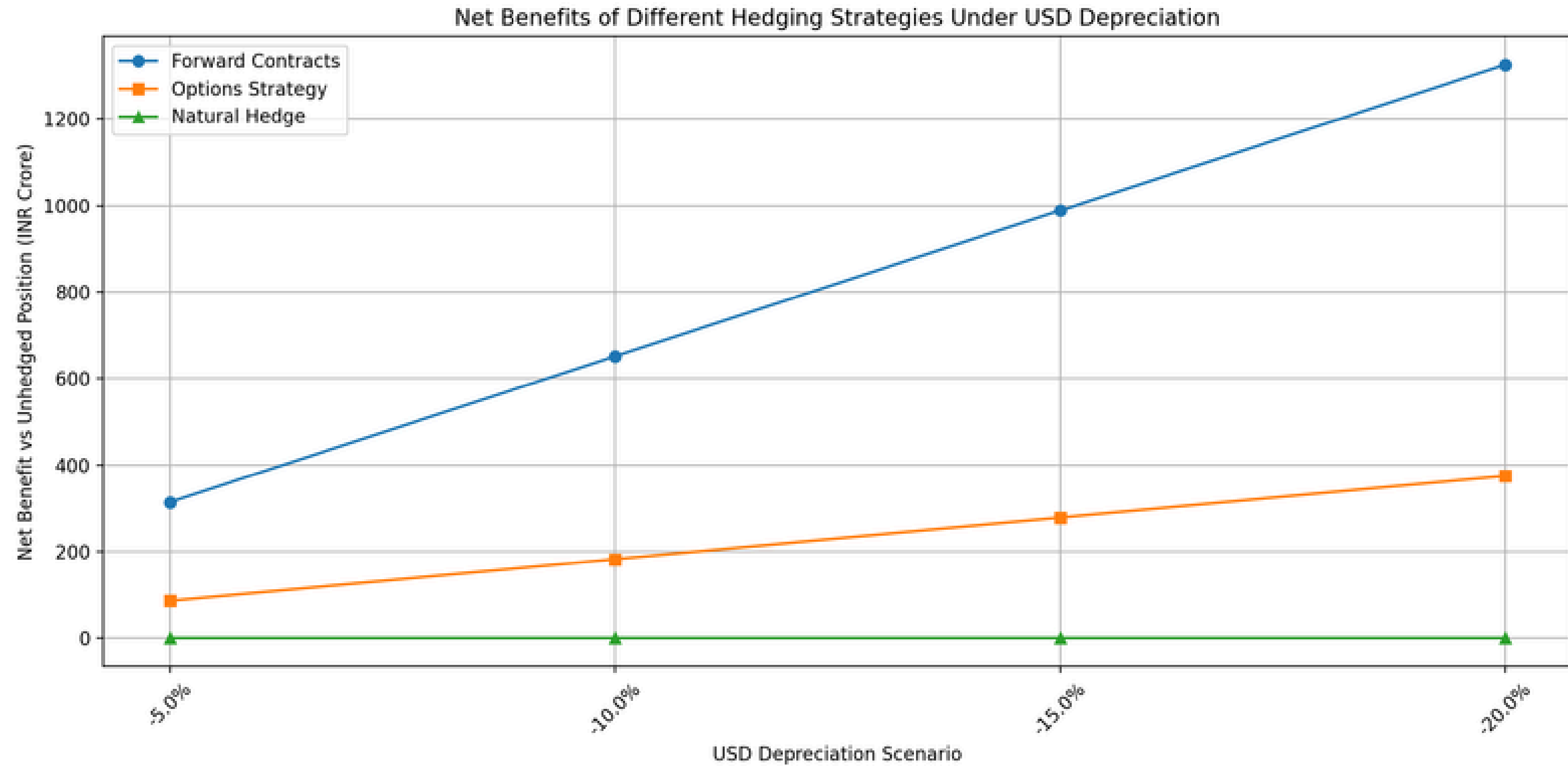


Scenario Analysis of XYZ Technologies in case of USD depreciation

Using same Hedging Strategy

	Depreciation	New USD/INR Rate	Unhedged Value	Forward Net Value	Forward Net Benefit	Options Net Value	Options Net Benefit	Natural Net Value	Natural Net Benefit
0	-5.00%	78.85	9146.6	9460.72	314.12	9232.44	85.84	9146.6	0
1	-10.00%	74.7	8665.2	9316.3	651.1	8847.32	182.12	8665.2	0
2	-15.00%	70.55	8183.8	9171.88	988.08	8462.2	278.4	8183.8	0
3	-20.00%	66.4	7702.4	9027.46	1325.06	8077.08	374.68	7702.4	0

Hedging Instruments



Regulatory Compliance



The Regulatory Compliance Overview highlights the frameworks and standards governing foreign exchange transactions, ensuring organizations adhere to legal requirements. It aims to enhance transparency, risk management, and accountability while safeguarding stakeholders' interests in the dynamic financial landscape

Regulatory Compliance Overview

REGULATION	DESCRIPTION
FEMA, 1999	The principal legal framework for regulating foreign exchange in India, overseen by the RBI.
Section 3, Section 6	Mandates risk reporting, including FX risks, requiring Board evaluation and disclosure of FX gains, losses, and hedging in financial statements.
IFRS 9	Specifies criteria for hedge accounting and the designation of hedging relationships.
RBI Notifications Under FEMA	Provides updates on permissible FX transactions and conditions for hedging.
Master Direction on Risk Management	Outlines regulations for hedging foreign exchange exposure.
Companies Act, 2013	Mandates detailed risk reporting, including foreign exchange risk, to ensure transparency.
Section 134	Requires the Board to evaluate FX risk and include this in the formal report.
Ind AS 109	Standards for the recognition, measurement, and disclosure of financial instruments.

Regulatory Compliance



This overview highlights how leading technology firms like TCS, Infosys, and Wipro exemplify best practices in regulatory compliance. By implementing comprehensive governance frameworks, adhering to AI regulations, and focusing on data protection, these companies serve as benchmarks for ensuring robust compliance and risk management in a rapidly evolving regulatory landscape.

TATA CONSULTANCY SERVICES

Governance, Risk, and Compliance (GRC)

TCS enhances compliance through comprehensive GRC services, integrating risk processes across operations.

AI Regulations

Committed to the EU AI Pact, TCS is preparing for compliance with the EU's AI Act for high-risk applications.

Cybersecurity Strategy

Ensures compliance with data protection regulations.

SOURCES: INFOSYS OFFICIAL WEBSITE, CONSULTANTS REVIEW

INFOSYS

Risk and Compliance Solutions

Infosys provides services in financial and operational risk, anticipating regulatory needs and defining governance policies.

AI Compliance

Also part of the EU AI Pact, Infosys promotes responsible AI usage and adherence to upcoming regulations.

Regulatory Reporting

Supports clients in meeting regulatory reporting requirements through GRC solutions.

SOURCES: INFOSYS OFFICIAL WEBSITE, CONSULTANTS REVIEW

WIPRO

Compliance Framework

Wipro adheres to international standards, ensuring compliance with local laws and effective risk management.

AI Regulations

Committed to the EU AI Pact, focusing on high-risk AI system regulations.

Data Privacy and Security

Aligns with global standards to protect client information in its compliance strategy.

SOURCES: WIPRO OFFICIAL WEBSITE, CONSULTANTS REVIEW

Monitoring and Reporting



Monitoring FX Exposure

“Real-time monitoring systems are critical for tracking fluctuations in foreign exchange rates and identifying potential risks.”
-FX Risk Management Best Practices for Technology Companies, Deloitte



Regular Reporting

“A structured reporting framework that provides regular updates to senior management is essential for managing FX exposure effectively.”
-Navigating FX Risk in High-Tech Companies, McKinsey & Company



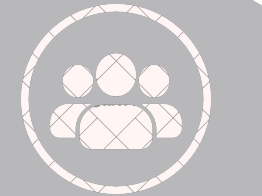
Dashboard Utilization

“Dashboards allow for the visual representation of FX risks and trends, enabling rapid insights and facilitating informed decision-making.”
-Enhancing FX Risk Management: A Framework for High-Tech Firms, PwC



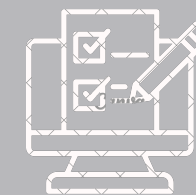
Stakeholder Engagement

“Engaging cross-functional teams in the reporting process ensures that all relevant perspectives are considered.”
-Integrating FX Management into Business Strategy, Boston Consulting Group.”



Compliance Monitoring

“Regular audits of monitoring processes are necessary to maintain compliance with local and international regulations.”
-Compliance and Risk Management for Technology Firms, EY”



Continuous Improvement

“Feedback mechanisms for refining monitoring and reporting processes enhance system efficiency and effectiveness.”
-FX Risk Management: A Continuous Improvement Approach, KPMG”



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EY, "Managing Foreign Exchange Risk: A Strategic Approach")



Strategy and Innovation stream

Thank You